EGG SEGMENTATION & TRACKING

Computer Vision Assignment | Technical Report

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SAM-based segmentation • Classical image processing • Evaluation • Video tracking

PROJECT REPOSITORY	github.com/ramritheesh/egg-segmentation-tracking
ANNOTATED VIDEO	Open tracking video on Google Drive
DATASET	Roboflow Egg Dataset — Version 1

PROJECT SUMMARY

This project uses the Segment Anything Model (SAM) to generate egg segmentation candidates, compares classical image-processing methods against SAM-derived pseudo-labels, and applies a contour-based detector with centroid tracking to a video. The report distinguishes measured agreement with pseudo-labels from verified ground-truth accuracy and documents tracking limitations.

Prepared for technical review and assignment submission

1. Executive Summary

The workflow covers four stages: (1) SAM-based image segmentation, (2) classical segmentation experiments, (3) quantitative evaluation on a held-out image subset, and (4) egg detection and tracking in a video. The selected classical method was GrabCut with mask initialization.

0.4739	0.5917	22 / 22	1,440
Mean IoU	Mean Dice	Eval images	Video frames

2. Dataset and Split

The Roboflow Egg Dataset, version 1, was stored locally under `data/raw/egg.v1i.folder/`. It contains 110 images at 640 × 640 resolution: 108 egg images and 2 unlabeled images. The split manifest is `outputs/task2/split_manifest.csv`.

Subset	Egg images	Unlabeled	Total
Development	86	1	87
Evaluation	22	1	23
Total	108	2	110

The split is approximately 4:1. An exact SHA-256 duplicate check found no exact duplicate leakage between subsets. Near-duplicate leakage was not tested and remains a limitation.

3. Task 1 — SAM-Based Segmentation

A pretrained SAM ViT-B checkpoint (`models/sam_vit_b.pth`) processed all 110 images. Candidate masks, selected masks, and overlays were saved under `outputs/task1_sam_batch/`. Configuration: `points_per_side=16`, `pred_iou_thresh=0.86`, `stability_score_thresh=0.92`, `crop_n_layers=0`, and `min_mask_region_area=100`.

The multi-egg candidate-selection pipeline was refined from v4 to v5. On 86 development egg images, v4 selected 201 candidates and had two zero-detection images. V5 increased the maximum candidate area ratio from 0.35 to 0.40, resulting in 203 selected candidates and one zero-detection image. It recovered a plausible candidate in `not_damaged_35`; `not_damaged_124` remained without a reliable selected candidate.

The selected masks are heuristic SAM-derived pseudo-labels. Candidate counts are not verified counts of true eggs. Visual inspection indicated plausible detections, but also possible false positives on hands or packaging and imperfect boundaries.

4. Task 2 — Classical Segmentation Experiments

Four methods were implemented: (1) HSV thresholding with morphology; (2) stricter HSV thresholding with connected-component filtering; (3) GrabCut initialized with a rectangle; and (4) GrabCut initialized with a foreground/background mask, followed by morphological opening and closing.

Development method	IoU	Dice	Precision	Recall
Experiment 1 — HSV	0.2986	0.4121	0.3201	0.7514
Experiment 2 — HSV components	0.1611	0.1881	0.2410	0.1747
Experiment 3 — GrabCut rectangle	0.3291	0.4109	0.3784	0.5285
Experiment 4 — GrabCut mask	0.3388	0.4127	0.3821	0.5142

Experiment 4 was retained as the provisional baseline because it achieved the highest development IoU and Dice. These development metrics used the earlier single-mask SAM references; the final evaluation used multi-instance v5 reference masks. The two evaluations are not directly interchangeable.

5. Task 3 — Final Quantitative Evaluation

The final evaluation processed all 22 evaluation images in the egg class. Task 1 v5 generated combined pseudo-reference masks, and Experiment 4 generated predictions. Results were saved under `outputs/task2/final_evaluation/`, including individual and summary CSV files and visual comparisons.

Metric	Mean result
Intersection over Union (IoU)	0.473918 (47.39%)
Dice coefficient	0.591723 (59.17%)
Precision	0.600877 (60.09%)
Recall	0.685487 (68.55%)
Evaluation coverage	22 of 22 images

High-agreement examples: `not_damaged_159` (IoU 0.9915), `not_damaged_22` (0.9443), and `not_damaged_162` (0.8139).

Low-agreement examples: `not_damaged_12` (0.0000), `not_damaged_34` (0.0000), `not_damaged_125` (0.1279), `not_damaged_128` (0.1860), and `not_damaged_121` (0.2201).

Failure analysis. In `not_damaged_12`, the masks had no overlapping foreground pixels and visual inspection showed that the methods selected different regions. In `not_damaged_34`, SAM v5 produced an empty reference mask while GrabCut predicted 86,068 foreground pixels; visual inspection suggested GrabCut captured the egg while SAM missed it. For `not_damaged_125` and `not_damaged_128`, relatively high recall but low precision indicated substantial predicted foreground outside the reference.

Interpretation: These metrics measure agreement with SAM v5 pseudo-labels, not true semantic accuracy against human pixel annotations. No human-annotated segmentation ground truth was available in this workflow.

6. Task 4 — Video Detection and Tracking

The supplied video was downloaded to `data/video/egg_tracking_video.mp4`. Properties: 1280 × 720 resolution, 24 FPS, 60.021 seconds, and approximately 1,440 frames. Detection used HSV thresholding for bright, low-saturation regions, morphological opening and closing, and contour filters based on area, bounding-box size, and aspect ratio. A test frame at approximately 10 seconds produced two egg detections.

The baseline tracker used centroid-distance matching with one-to-one greedy assignment and a temporary disappearance allowance. The baseline processed all frames and assigned 10 track IDs. Outputs: `outputs/task4/egg_tracking_output.mp4` and `outputs/task4/egg_tracking_log.csv`.

A second experimental version used linear motion prediction from the last two observed centroids and assigned 15 IDs. It performed worse by this count and was not retained as the improvement. V2 outputs were `egg_tracking_output_v2.mp4` and `egg_tracking_log_v2.csv`.

> **Tracking limitation**
> Hands repeatedly occlude eggs, causing the detector to lose or fragment egg regions. Reappearing eggs may receive new IDs. Ten assigned IDs does not mean ten distinct physical eggs. No manually annotated video tracks or identity-switch benchmark was available, so identity accuracy has not been quantitatively validated.

Annotated output video: View egg_tracking_output.mp4 on Google Drive

7. Tools, Reproducibility, and Outputs

Tools: Python 3.13.4, OpenCV, PyTorch, Segment Anything (SAM) ViT-B, NumPy, Pandas, FFmpeg/FFprobe, and yt-dlp.

Key outputs: Task 1 — `outputs/task1_sam_batch/`; split manifest — `outputs/task2/split_manifest.csv`; Task 2 development metrics — `outputs/task2/evaluation/`; final metrics — `outputs/task2/final_evaluation/`; Task 4 baseline video and CSV — `outputs/task4/`.

Execution commands

```
python3 src/task1_batch_sam.py
python3 src/task2/run_dev_experiments.py
python3 src/task2/evaluate.py
python3 src/task4/track_eggs.py
```

The repository README contains setup and execution information. The SAM checkpoint and dataset are not committed to GitHub; users must obtain them separately and place them in the documented locations.

8. Limitations and Future Work

Recommended next steps are: (1) create human-annotated masks for a representative subset; (2) audit SAM false positives and false negatives; (3) test near-duplicate leakage; (4) improve detection under hand occlusion; (5) evaluate a standard motion model with global assignment, such as Kalman filtering with Hungarian assignment; and (6) calculate ID switches, fragmentation, and IDF1 only after manually annotating video tracks.

9. Conclusion

The project completed a SAM-based image segmentation pipeline, compared four classical segmentation methods, evaluated GrabCut on 22 evaluation images, and generated an annotated video using an OpenCV tracking baseline. The final image evaluation achieved mean IoU 0.4739 and Dice 0.5917 against SAM v5 pseudo-labels.

The analysis identified cases where pseudo-reference failures affect metric interpretation. The video pipeline processed all 1,440 frames, but hand occlusion and detection fragmentation caused track-ID inconsistencies.

10. Submission Links and AI-Use Disclosure

Public GitHub repository: github.com/ramritheesh/egg-segmentation-tracking

Google Drive annotated video: Open egg_tracking_output.mp4

Dataset: Roboflow Egg Dataset — Version 1

Source video: Raw and Boiled Egg Spinning — Cool Science Experiment

AI-use disclosure: Generative AI assistance was used for code drafting, debugging guidance, experiment interpretation, and report preparation. The author should review and verify the final implementation, outputs, and claims. Update this disclosure if additional AI tools or services were used.